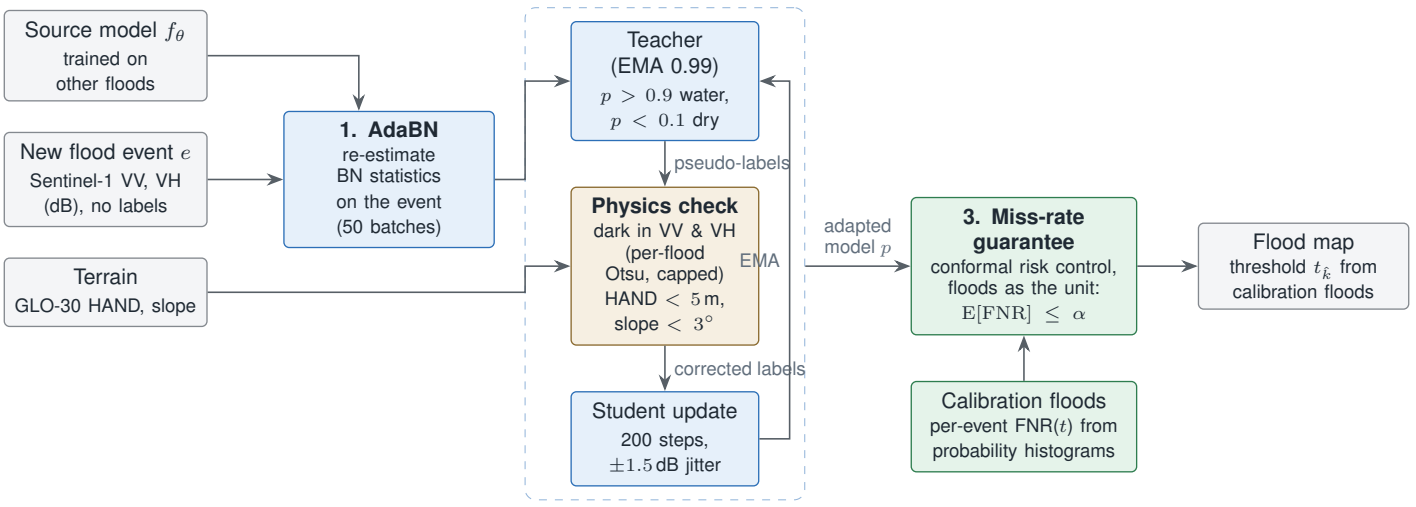

2. Physics-checked self-training
Source model f_θ
trained on
other floods
New flood event e
Sentinel-1 VV, VH
(dB), no labels
Terrain
GLO-30 HAND, slope
1. AdaBN
re-estimate
BN statistics
on the event
(50 batches)
Teacher
(EMA 0.99)
$p > 0.9$ water,
$p < 0.1$ dry
pseudo-labels
Physics check
dark in VV & VH
(per-flood
Otsu, capped)
HAND < 5 m,
slope $< 3°$
EMA
corrected labels
Student update
200 steps,
± 1.5 dB jitter
adapted
model p
3. Miss-rate
guarantee
conformal risk control,
floods as the unit:
$\mathrm{E}[\mathrm{FNR}] \leq \alpha$
Calibration floods
per-event FNR(t) from
probability histograms
Flood map
threshold $t_{\hat{k}}$ from
calibration floods